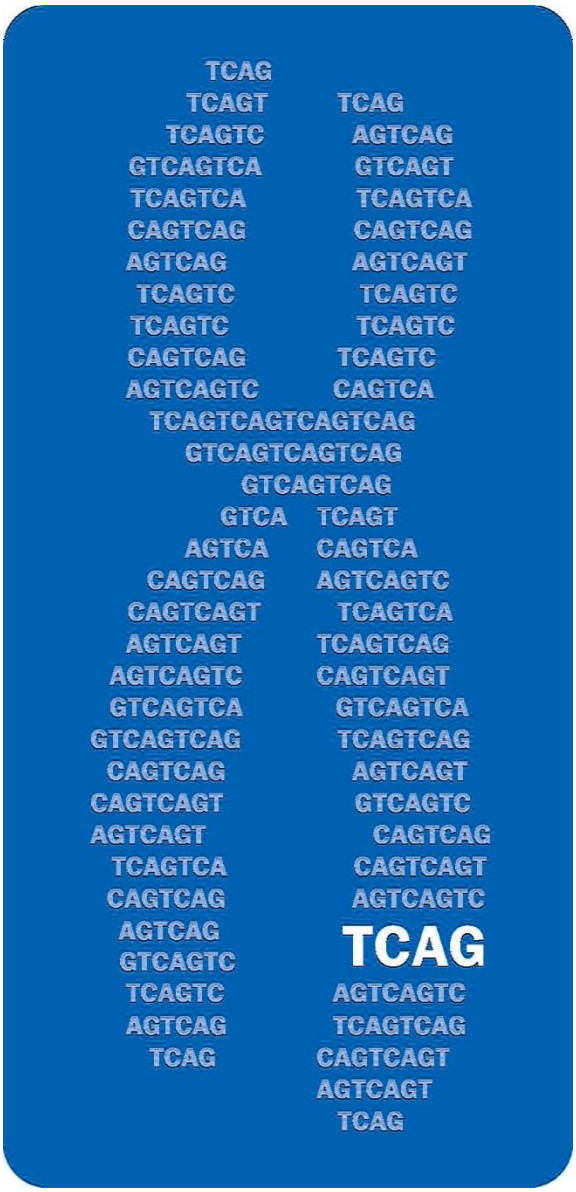
IEI-BISSH project

2026-10-16 gene list supplementary report



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Data overview

The IEI-BISSH (IEI: inborn errors of immunity; BISSH: bacterial immune system structural homology) data associated with this report are stored in `2026-10-07_for_c_marshal.tsv`. Each row gives a single bacteria-to-human protein pair, where the proteins in each pair exhibit structural homology. Bacterial proteins are drawn from the DefenseFinder dataset and therefore have a documented role in the immune systems of bacteria.

Our pipeline leverages the FoldSeek all-v-all structural similarity network to identify human proteins that have structural homology with DefenseFinder proteins. Pipeline hits may play a role in the human immune system, particularly antiviral response, and may be used as evidence in a larger clinical genomics discovery workflow.

Protein pairs were filtered for $composite\ score \geq 0.85$, indicating strong evidence for structural homology. For each unique human UniProt accession, only the highest-scoring bacteria-human protein pair is included. The UniProt accessions map to 720 distinct human gene names.

These data were generated by The Centre for Applied Genomics (TCAG), Chantel Trost lab, and the Centre for Computational Medicine (CCM) at the Hospital for Sick Children. Important columns are described below:

Column name	Column description
composite_score	Overall in-house evidence score for structural homology out of 1.0. Incorporates the FoldSeek e-value, PyMOL-super root-mean-squared deviation (RMSD), TM-scores for each protein, and AlphaFold pLDDTs for each protein
defense_uniprot_ac	UniprotKB accession for the bacterial protein
human_uniprot_ac	UniprotKB accession for the human protein
human_gene_in_griid	Boolean value indicating whether the human protein maps to a gene on the Genetic Research in Inflammatory and immune Disorders (GRIID) gene panel
human_gene_in_IUIS	Boolean value indicating whether the human protein maps to a gene on the International Union of Immunological Societies (IUIS) IEI expert committee's 2025 gene panel
blastp_*	BLASTp pairwise alignment results for the bacterial and human protein sequences; these annotations give evidence for sequence homology. Note that BLASTp e-values and bitscores are stronger than typical BLASTp results as the alignment is pairwise. BLASTp $evalue \leq 10^{-3}$ is the typical permissive threshold for significant sequence homology.
marshall_immune_subpanel	

Composite scoring function

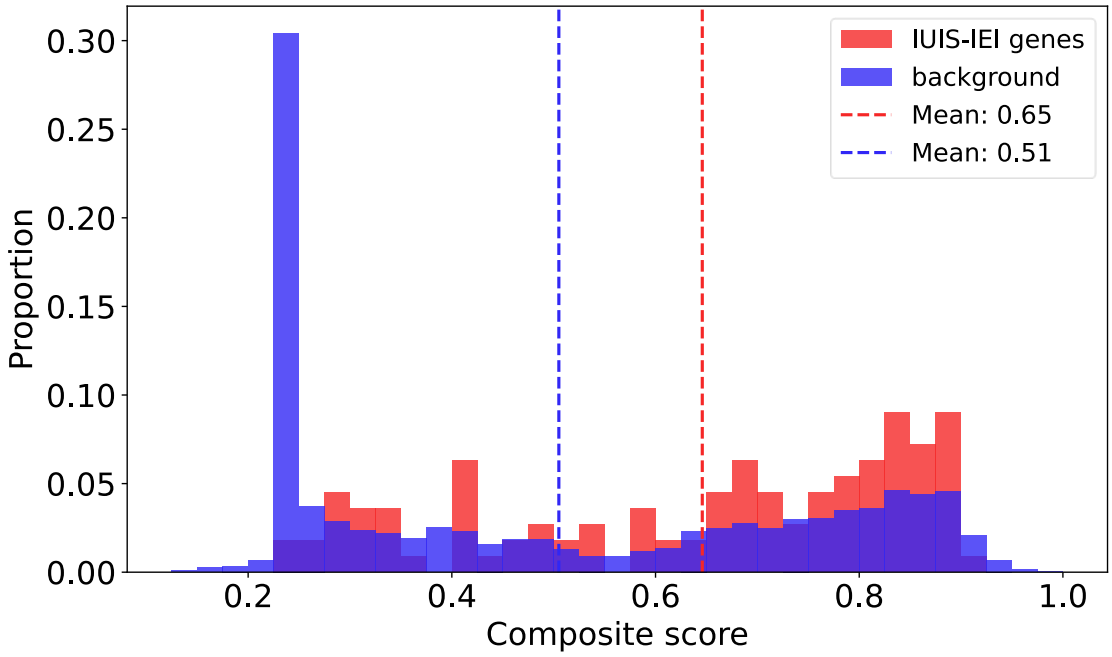
The composite score for each pair by the following parameters. Each individual term is between 0.0 and 1.0 and is incorporated into a weighted sum that in turn yields a final score between 0.0 and 1.0:

Composite score

0.0 → 1.0

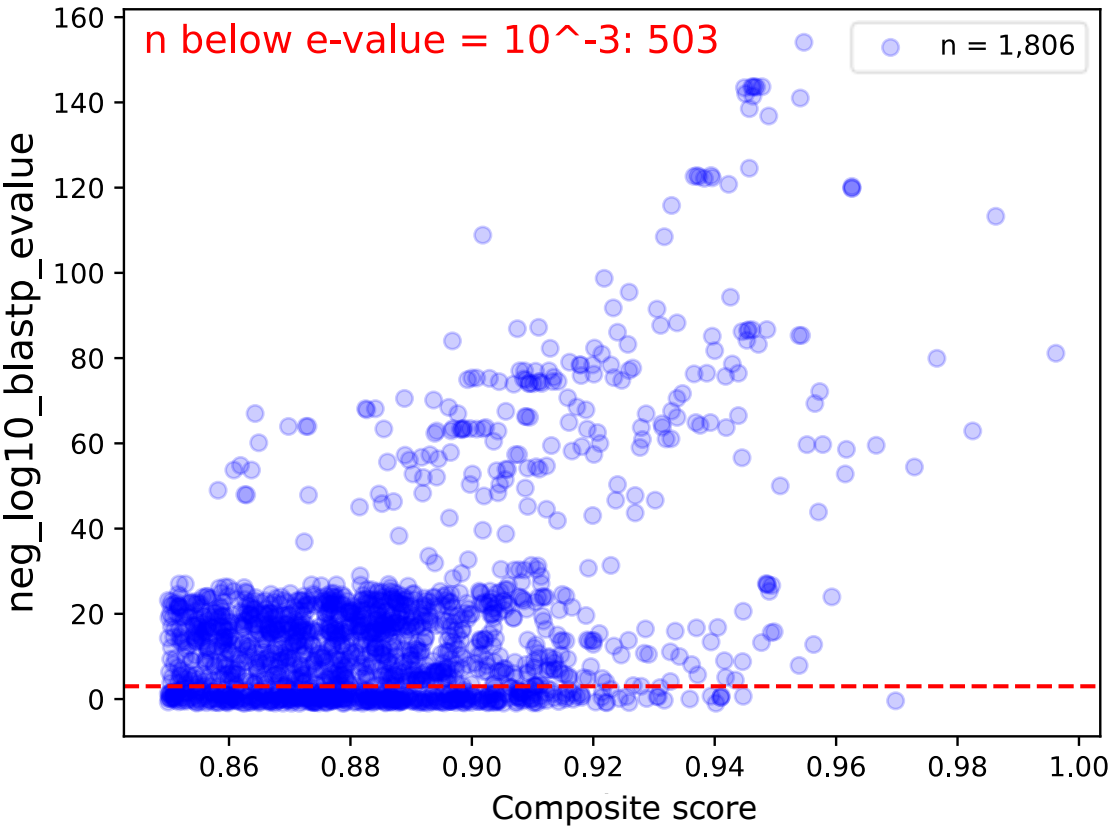
0.40	× PyMOL score term	$= \frac{1.06}{1 + e^{0.635(PyMOL_RMSD - 4.94)}}$
0.35	× FoldSeek e-value score term	$= 1 - e^{-\left(\frac{-\log_{10}(e_value)}{4.7}\right)^{1.03}}$
0.05	× TM score (human protein)	
0.05	× TM score (bacterial protein)	
0.075	× pLDDT (human protein) / 100	
0.075	× pLDDT (bacterial protein) / 100	

Figure 1: composite score distribution in IUIS-IEI proteins and background



Of the 591 human genes in the IUIS-IEI expert committee's gene panel, 148 are included among the IEI-BISSH pipeline returns. Those what were filtered out by the pipeline do not have a FoldSeek $evalue \leq 0.05$ with a DefenseFinder protein. The mean composite score for protein pairs that map to IUIS-IEI genes is 0.14 greater than background.

Figure 2: 27.9% of high-scoring pipeline hits exhibit insignificant sequence homology



Each point represents a human-bacterial protein pair in the high-scoring ($composite_{score} \geq 0.85$) dataset. Darker colour indicates overlapping points. BLASTp alignment fails for 42 of the 1,849 protein pairs. 503 if the 1,806 (27.9%) valid BLASTp returns fall below the BLASTp $evalue \leq 10^{-3}$ significance threshold, suggesting the IEI-BISSH pipeline recovers conservation relationships that cannot be detected by sequence homology alone.